# Human Values in Obituaries: A Computational Analysis of Memorial Texts

Jiaxin Li

**Abstract**

Obituaries compress a life into a short public narrative. The people, events, relationships, and personal qualities selected for inclusion therefore provide a useful record of what a society chooses to remember. This project studies value expression in 379 *New York Times* obituaries and uses Touché23-ValueEval to build a ten-category human-value classifier. The analysis combines sentence-level text processing, TF-IDF features, one-vs-rest logistic regression, an interpretable lexicon check, value co-occurrence analysis, and decade-level aggregation. The project produces three connected views of memorial language: which values are most visible, which values tend to appear together, and how the strongest value signals vary across time. The resulting workflow turns a qualitative question about remembrance into a compact computational study while keeping the original sentences available for interpretation.

## 1 Introduction

An obituary is both a biographical record and a selective account of a life. Within limited space, a writer chooses which achievements, relationships, responsibilities, beliefs, and personal qualities deserve emphasis. When many obituaries are examined together, these choices reveal recurring patterns in how lives are remembered and socially evaluated. This potential has been demonstrated at a much larger scale by Markowitz et al., who analyzed up to 38,245,928 U.S. obituaries published between 1998 and 2024 through the lens of Schwartz's theory of basic human values [1]. Their study suggests that obituary language contains systematic signals about the values emphasized in memorial writing. Motivated by this perspective, I examine which values are most prominent in obituary language, which tend to appear together, and how their expression changes across historical periods.

The core corpus contains 379 *New York Times* obituaries published between 1852 and 2007. These texts cover prominent public figures across a wide range of occupations and span more than a century, providing a compact corpus for studying memorial language across social roles and historical contexts. I interpret the model outputs as properties of the text rather than direct measurements of the deceased individuals' personal beliefs.

The project combines digital humanities, computational social science, and natural language processing to examine an abstract question about what kinds of lives are presented as worth remembering through observable textual patterns. Descriptions of professional distinction may reflect achievement, references to family care may express benevolence or security, and discussions of faith, custom, or civic responsibility may signal tradition, conformity, or universalism. Because each prediction remains linked to a source sentence, aggregate patterns can be traced back to concrete language and interpreted in context. This framework provides a focused setting for combining text processing, machine learning, statistical analysis, and close reading within a single research question.

## 2 Methods

### 2.1 Data and analytical roles

The project uses two main datasets. The obituary corpus provides the target texts for cultural analysis. Touché23-ValueEval provides labeled arguments for learning value-related language [2, 3]. The official

training and validation splits are kept separate so that model performance is evaluated on unseen labeled texts before the classifier is applied to the obituaries.

The twenty fine-grained ValueEval labels are consolidated into ten broader categories: self-direction, stimulation, hedonism, achievement, power, security, conformity, tradition, benevolence, and universalism. The broader categories reduce sparsity and provide a level of interpretation suitable for the obituary corpus.

### 2.2 Text representation and value classification

Each obituary is stored at both document and sentence level. The document record retains the subject name, death year, full text, and length. Sentence records retain document membership, position, year, and text. This structure supports corpus-level statistics while preserving the evidence needed for close reading. For ValueEval, the premise and conclusion are concatenated as the input text. A TF-IDF representation uses unigrams and bigrams, and a one-vs-rest logistic regression model estimates each of the ten value categories independently. For value $v$ and sentence $x$, the classifier estimates

$$P(y_v = 1 \mid x) = \sigma\Big(w_v^\top \phi(x) + b_v\Big), \tag{1}$$

where $\phi(x)$ is the TF-IDF feature vector and $\sigma$ is the logistic function. This formulation allows several values to be active in the same sentence.

A small value lexicon provides a second, transparent signal. For a sentence $x$, its lexicon score is

$$S_v^{\text{lex}}(x) = 1000 \times \frac{C_v(x)}{N(x)}, \tag{2}$$

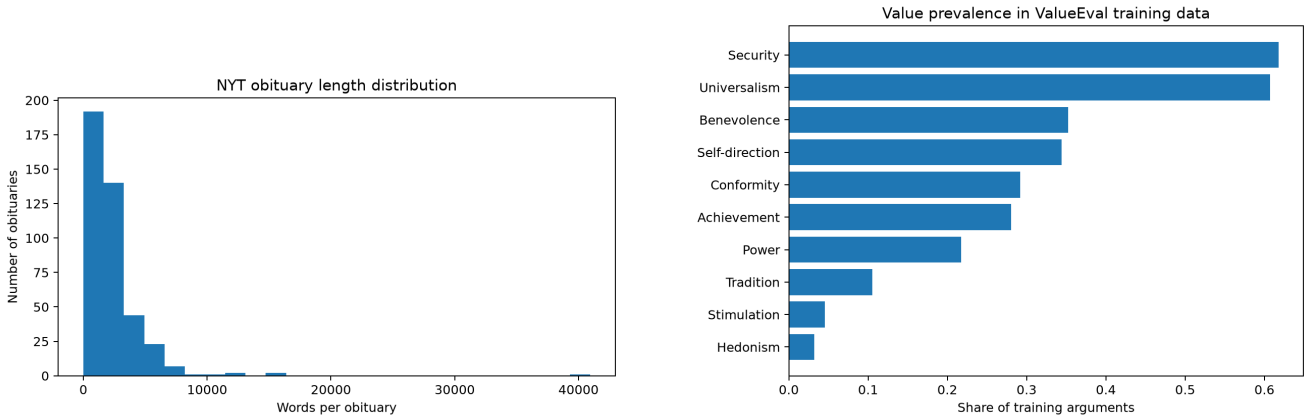
where $C_v(x)$ is the number of value-related term matches and $N(x)$ is the sentence length. The lexicon is used as an interpretive check rather than a replacement for the classifier. Model quality is summarized with micro-F1, macro-F1, weighted-F1, and class-specific precision, recall, and F1. After validation, the classifier assigns ten probabilities to every obituary sentence. Mean probability describes overall value prominence; a threshold of 0.5 defines sentence-level co-occurrence; decade averages summarize longer-run changes.

Together, these measures support three complementary analyses of the obituary corpus: overall value prominence, sentence-level relationships among values, and changes in value signals across decades.

## 3 Results

### 3.1 Corpus structure and training-label balance

Figure 1 presents two diagnostics that shape the later interpretation. The obituary-length distribution shows how much the amount of available biographical detail varies across subjects. The ValueEval prevalence plot shows that the ten target categories are not equally represented in the training data. This imbalance motivates class-balanced logistic regression and makes per-category F1 important alongside the aggregate scores.




**(a)** Obituary length distribution. **(b)** Value prevalence in the training set.

**Figure 1:** Corpus scale and supervision balance.

## 3.2 Value-classification performance

The classifier reaches a micro-F1 of 0.580, a macro-F1 of 0.447, and a weighted-F1 of 0.572 on the ValueEval validation split. Table 1 reports performance separately for each value. The strongest category is Security with F1 = 0.755, while Stimulation has the lowest F1 at 0.117. The gap between categories is important because all later obituary probabilities inherit the classifier's uneven ability to recognize different kinds of value language.

**Table 1:** Validation performance by value category.

| Value | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Security | 0.705 | 0.813 | 0.755 | 1160 |
| Universalism | 0.657 | 0.689 | 0.673 | 1100 |
| Benevolence | 0.560 | 0.627 | 0.592 | 805 |
| Achievement | 0.491 | 0.590 | 0.536 | 575 |
| Self-direction | 0.516 | 0.521 | 0.519 | 610 |
| Conformity | 0.488 | 0.410 | 0.446 | 568 |
| Power | 0.356 | 0.376 | 0.366 | 290 |
| Tradition | 0.364 | 0.209 | 0.266 | 172 |
| Hedonism | 0.196 | 0.214 | 0.205 | 103 |
| Stimulation | 0.153 | 0.094 | 0.117 | 138 |

Figure 2 makes this variation easier to compare. Categories with stronger F1 can support more confident corpus-level interpretation, while weaker categories should be read more cautiously and checked against the underlying sentences and lexicon signal.

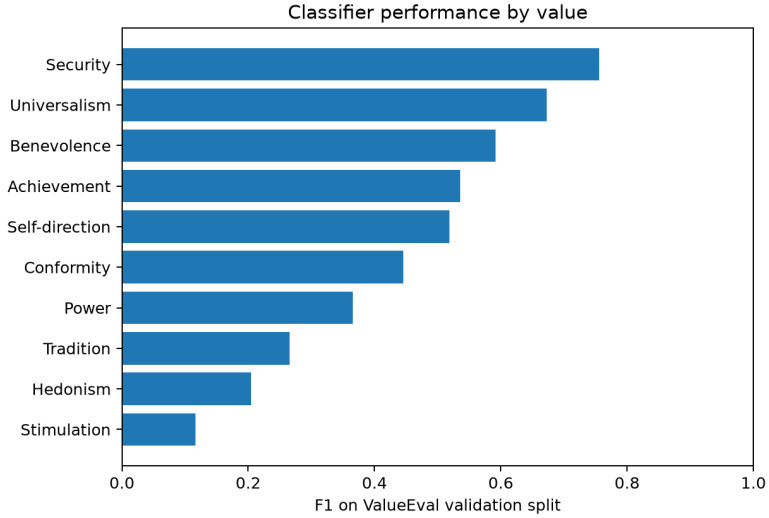



**Figure 2:** F1 score for each of the ten value categories on the ValueEval validation split.

### 3.3 Value signals in the obituary corpus

After validation, the model is applied to every obituary sentence. Table 2 combines two views of each value: the average classifier probability and the share of sentences containing at least one lexicon term for that category. The highest mean predicted probability is observed for Universalism (0.557), followed by Security (0.524). These rankings describe what the obituary language emphasizes most strongly under the learned value representation.

**Table 2:** Value signals in the obituary corpus.

| Value | Mean probability | Lexicon coverage |
|---|---|---|
| Universalism | 0.557 | 0.037 |
| Security | 0.524 | 0.090 |
| Benevolence | 0.465 | 0.036 |
| Achievement | 0.456 | 0.088 |
| Self-direction | 0.434 | 0.011 |
| Conformity | 0.434 | 0.021 |
| Power | 0.392 | 0.151 |
| Tradition | 0.243 | 0.024 |
| Stimulation | 0.225 | 0.009 |
| Hedonism | 0.137 | 0.042 |

Figure 3 shows the same classifier-based ranking visually. The lexicon column in Table 2 provides a useful comparison: agreement between probability and lexicon coverage suggests a value is often expressed with explicit vocabulary, while disagreement indicates that the classifier may be relying on broader phrasing learned from ValueEval. This comparison keeps the aggregate ranking connected to interpretable textual evidence.

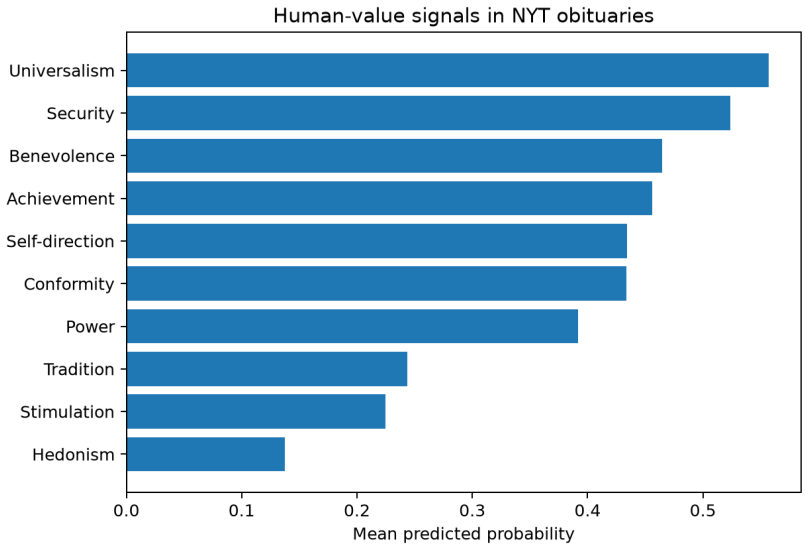
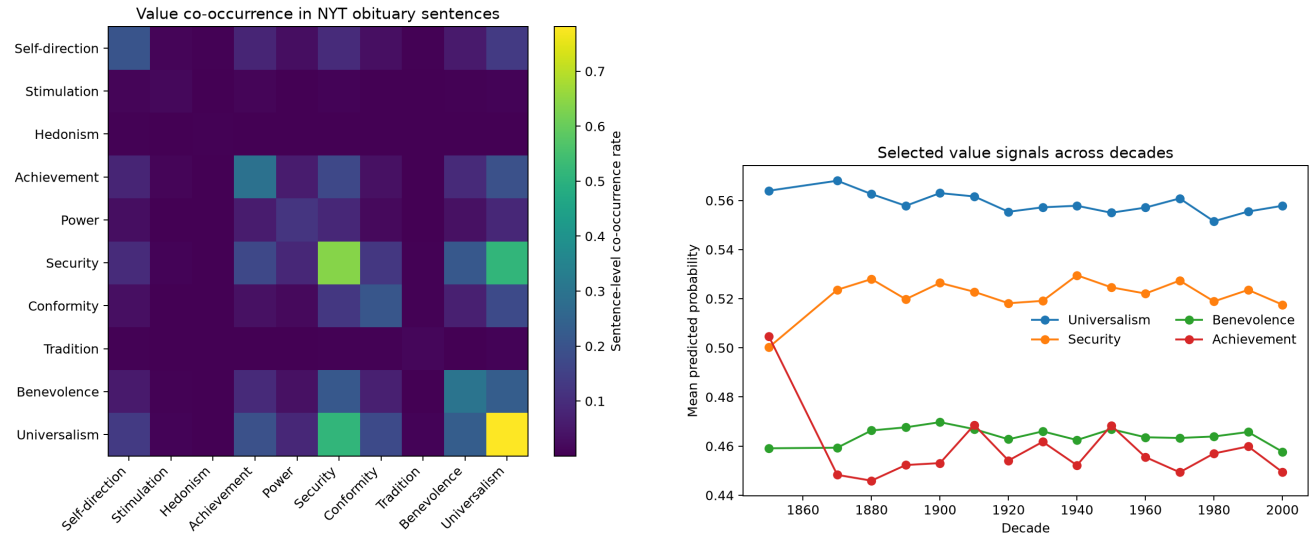



**Figure 3:** Mean predicted value probabilities across obituary sentences.

### 3.4 Value combinations and historical variation

The co-occurrence matrix in Figure 4a measures how often two values both exceed the 0.5 probability threshold in the same sentence. The strongest off-diagonal pair is Security with Universalism, with a sentence-level co-occurrence rate of 0.515. This view moves beyond a simple ranking by showing how memorial narratives combine values within the same statement.

The decade plot in Figure 4b follows the four most prevalent predicted values through time. Among these values, Achievement shows the largest absolute change between the 1850s and the 2000s, moving from a mean probability of 0.505 to 0.449. The curve describes change within this particular obituary collection; it is most useful as a guide for returning to texts from different periods and examining how the language of remembrance shifts.




**(a)** Sentence-level value co-occurrence. **(b)** Selected value signals by decade.

**Figure 4:** Relationships among values and their historical variation in the obituary corpus.

## 4 Conclusion

This project develops a compact computational framework for studying what memorial writing chooses to emphasize about a life. A social-science value taxonomy provides the conceptual structure, a multi-label text classifier turns sentences into comparable value signals, and corpus analysis summarizes prominence, co-occurrence, and historical movement. The design keeps the statistical layer tied to source sentences, so patterns can be interpreted through the language that produced them.

The results provide three complementary views of obituary writing. The value distribution identifies the themes most strongly represented in the corpus. The co-occurrence analysis shows how values are combined within individual statements. The decade analysis places those signals in historical order and reveals which forms of value expression change most across the collection. Together, these outputs demonstrate how a relatively small corpus can support a meaningful study of public memory when computational measures and textual interpretation are used together.

## References

[1] D. M. Markowitz, T. Mazzucchi, S. Syropoulos, K. F. Law, and L. Young. An exploration of basic human values in 38 million obituaries over 30 years. *Proceedings of the National Academy of Sciences*, 122(35):e2510318122, 2025. DOI: 10.1073/pnas.2510318122.

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[4] D. Grangier et al. Wikipedia Biography Dataset. https://github.com/DavidGrangier/wikipedia-biography-dataset.